

Multi-Agent System for Real-Time Microgrid Management and Multi-Objective Optimization Incorporating Explainable Forecasting Models

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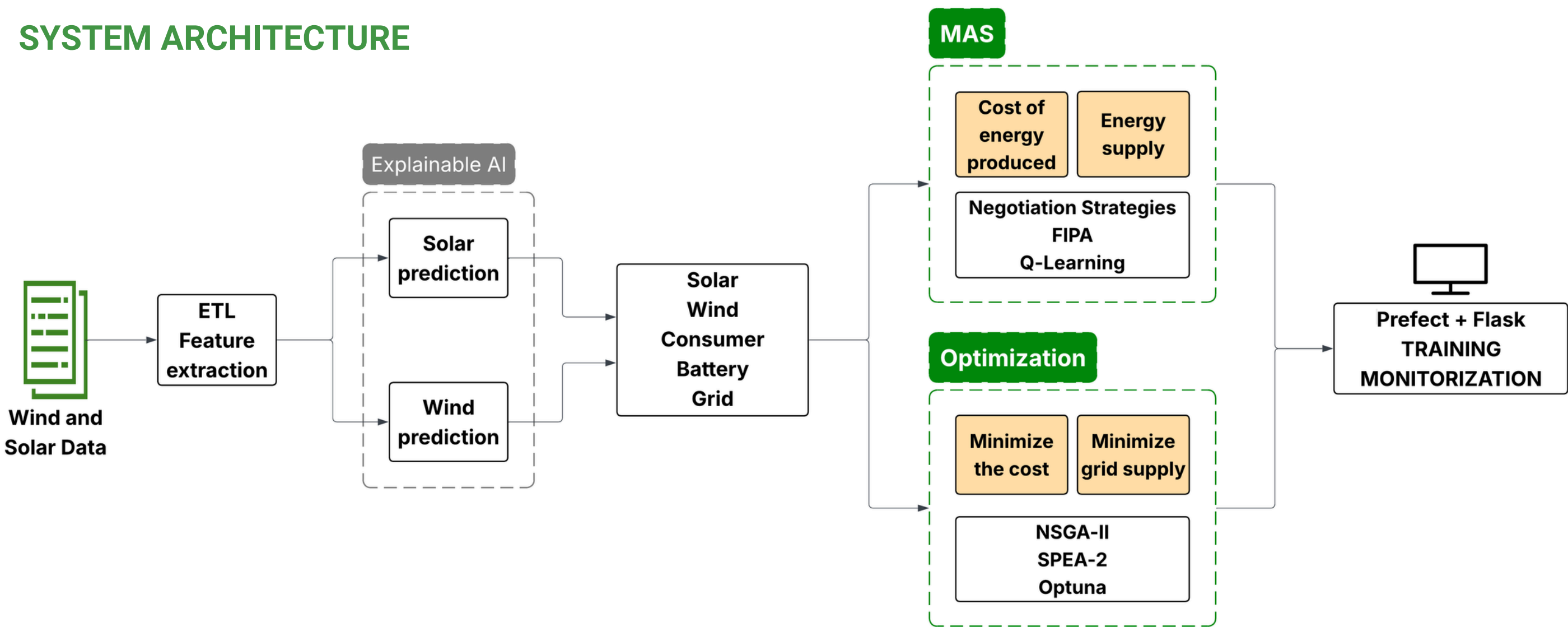
MOTIVATION

Renewable microgrids must operate under uncertain solar and wind generation while balancing operational cost and grid dependency.

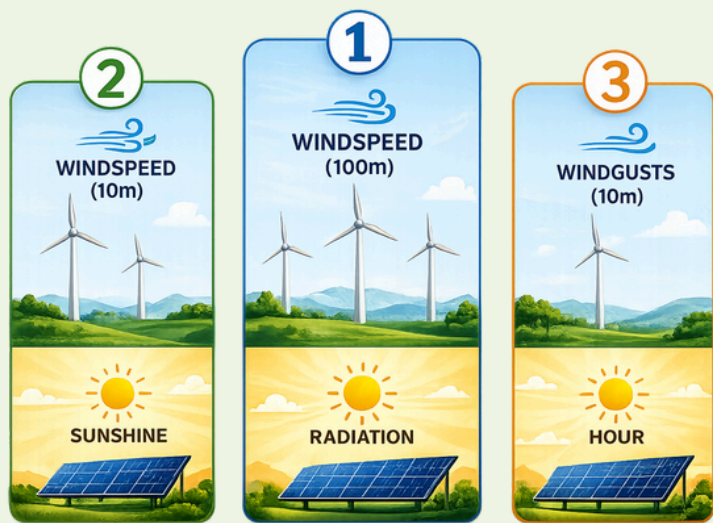
This work proposes two different solutions for real-time energy management.



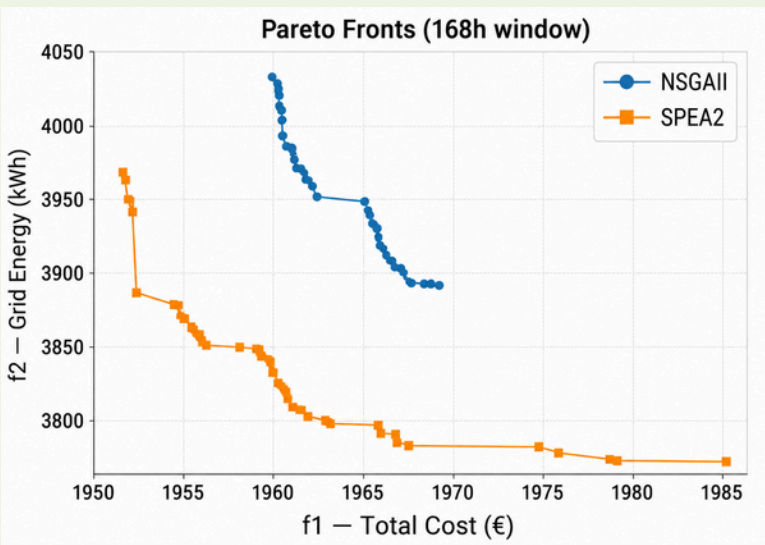
SYSTEM ARCHITECTURE



RESULTS



ASPECT	COMPETITIVE	COOPERATIVE	NEGOTIATION
OBJECTIVE	Maximize individual profit while avoiding penalties	Maximize overall microgrid welfare	Maximize profit and market share
BEHAVIOUR	Honest	Honest / Hide	Honest
TOTAL REWARD (€)	Medium	Lowest	HIGHEST
GRID DEPENDENCY (kWh)	LOWEST	Medium	Highest
LEARNING STYLE	Stable profit-driven policy	Most conservative behaviour	Most strategic and aggressive learning
STATE REPRESENTATION	27	27	81



CONCLUSIONS

XAI identified the **most influential variables** for renewable generation forecasting.

$$P = \frac{1}{2} \rho A v^3, P_{PV} = \eta A G$$

Selected predictors are consistent with the **governing physical equations**.

Negotiation achieved the **highest cumulative reward**.

Honest declarations consistently outperformed deceptive behaviours.

Dynamic penalties reduced over-commitment and improved grid reliability.

SPEA2 better than NSGA-II.

Better **Pareto-front quality** at the expense of higher **computational time**.

REFERENCES

- [1] J. F. Manwell, J. G. McGowan, and A. L. Rogers, Wind Energy Explained: Theory, Design and Application, 2nd ed. Hoboken, NJ, USA: Wiley, 2010.
- [2] R. Messenger and J. Ventre, Photovoltaic Systems Engineering, 2nd ed. Boca Raton, FL, USA: CRC Press, 2010.